Supply Chain Analytics 11 (2025) 100130

Contents lists available at ScienceDirect

Supply Chain Analytics

journal homepage: www.sciencedirect.com/journal/supply-chain-analytics

An analytical review of predictive methods for delivery delays in supply chains

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ARTICLE INFO

Keywords:
Predictive analytics
Supply chain forecasting
Machine learning models
Delivery delay prediction
Data-driven decision
Logistics optimization

ABSTRACT

Predicting delivery delays is crucial for companies, especially in times of increasing global uncertainty and vulnerable supply chains. Machine learning (ML) offers significant potential to improve the forecast performance and quality of delivery delay prediction. Although various prediction approaches have been proposed in research, a structured and comprehensive overview is lacking. This paper addresses this gap by conducting a systematic literature review on the direct prediction of delivery delays. The objective is to identify applied prediction approaches and data sources, assess their readiness for real-world implementation, and derive a research agenda. The findings reveal that current research often focuses on marginal optimization of prediction performance while lacking practical applicability. Furthermore, most studies emphasize classifying deliveries as on time or delayed, rather than predicting the actual delay magnitude. Regarding the data used for prediction, combining enterprise resource planning (ERP) data with data from logistics improves prediction performance. However, environmental and location data, which could be easily integrated into ERP-based ML models, are rarely considered. This indicates a misalignment in current research, emphasizing the need for models combining practical applicability with predictive accuracy. Further research is required to address these identified deficits. Therefore, the present paper proposes a research agenda, to prioritize the most important deficits. These include, among others the industrial application, optimal prediction timing and ideal data combinations to achieve high prediction accuracy. It also highlights the need for integrated decision support systems that provide prediction-based recommendations, enhancing the practical value of predictive models in supply chain management.

1. Introduction

In manufacturing companies, production often relies on the timely delivery of parts and components from a global and increasingly complex network of external suppliers [1,2]. Disruptions in the supply chain, whether due to delays, quality issues, or other factors, can result in production stoppages, increased costs, and missed customer deadlines [3]. This risk is especially high in just-in-time environments, where companies rely on tightly synchronized deliveries from multiple independent suppliers [4]. In this case, production and therefore the entire value chain can collapse due to a single missing component. To manage such risks and take appropriate countermeasures, companies can rely on proactive or reactive strategies [5]. In the context of supply chain management, proactive strategies are recommended to minimize risks before they impact production schedules and delivery deadlines. One proactive risk management strategy to ensure a timely start to production is the prediction of potential delays in the supply chain. Developments in the field of data analysis using artificial intelligence (AI), in particular machine learning (ML), and the availability of large data sets in the supply chain offer many approaches and methods for predicting delivery delays as part of proactive risk management [6–8].

Given the wide variance of different algorithms used in various approaches to implement these predictions, along with the use of equally diverse datasets containing different data classes, a clear and comprehensive presentation of the current state of the art in delivery delay prediction is essential to provide user guidance. Furthermore, the relevance of these prediction methods for industrial applications, resulting from the large processing power and variance of ML models, highlights the need for such an overview. While existing literature reviews confirm the thematic relevance of delivery delay prediction, to the best of the authors' knowledge, no publication fully addresses this topic or consolidates the current state of research.

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https://doi.org/10.1016/j.sca.2025.100130
Received 21 January 2025; Received in revised form 19 May 2025; Accepted 20 May 2025
Available online 22 May 2025
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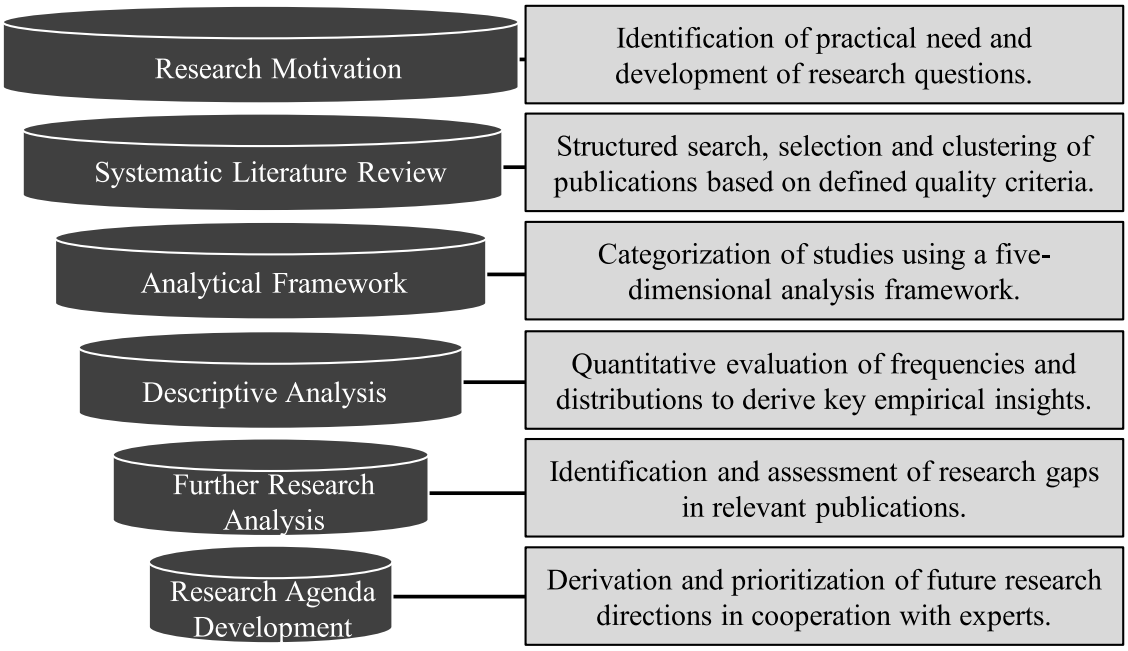

Fig. 1. Methodological approach used in this paper, visualized as taxonomy framework.

This gap not only emphasizes the need to close this deficit by presenting a structured and consolidated overview but also raises the question of how future research in this field can be systematically aligned to address open questions. Therefore, the overarching research question that underlies this paper is: What is the current research state for delivery delay prediction, and how can current research gaps be systematically consolidated into a research agenda? To answer the question, close this gap and establish a clear direction for future studies, the development of a research agenda is recommended to direct the future focus of research. A research agenda forms an overarching plan that provides a problem-solving framework for achieving research objectives and for generating knowledge in a targeted manner based on a series of specific and detailed measures [9,10]. The agenda thus serves as a central tool for the strategic orientation of research and innovation by consolidating and structuring the open research topics and questions in a concrete roadmap with milestones for future studies [11]. Resulting, the agenda contributes significantly to the generation of knowledge and the further development of the research field [11].

Addressing the identified need for a structured and consolidated review of delivery delay prediction, this paper aims to achieve two main objectives: First, a systematic literature review (SLR) is conducted to consolidate the current state of research and identify existing gaps in delivery delay prediction. This analysis provides insights into the methods, data sources, and validation approaches currently used for delivery delay prediction. Secondly, based on the results of the SLR a research agenda will be developed, which provides a strategic roadmap for future studies. The agenda consolidates and structures open research questions (RQ), enabling targeted knowledge generation and innovation in this research area. The methodological approach of the paper is also presented in Fig. 1 below in the form of a taxonomy framework, which summarizes the individual steps with a brief explanation of the objective.

To answer the overarching research question, four specific and more detailed research questions will be presented and discussed in detail in Section 3. These questions provide a deeper examination of the core aspects of delivery delay prediction, including prediction approaches, data sources, validation methods, and implications for future research. This paper makes the following two main contributions to the extant literature:

It shows that a large proportion of publications only use classification algorithms for prediction and therefore do not determine a specific delay duration. It also shows that predictions based on several data classes are more accurate. In particular, the use of location-based supplier data is an important input in addition to order and material data.

The research agenda encompasses eight research areas that emerged from the literature analysis and were prioritized by experts using pairwise comparison. Exemplary the integration of existing prediction models into the manufacturing industry and the development of a decision support system (DSS) to generate recommendations for actions based on the prediction are key aspects of this agenda.

Therefore, this paper is structured as follows. The next section introduces the concepts of delivery time and delivery delay and provides a distinction of this SLR from existing reviews in the context of supply chain management (SCM). Section 3 presents the methodological approach used to identify, select and analyze the relevant publications and provides the detailed research questions. Section 4 develops a framework based on the relevant publications analyzed, which is used to conduct a descriptive analysis to answer the defined RQ based on the analysis results. Section 5 addresses the main research gaps identified in the literature and proposes an agenda for future research in predicting delivery delays. Section 6 outlines the limitations of this paper and Section 7 summarizes the findings and contributions of this paper.

2. State of the art: delivery time and delivery time prediction

By definition, the term delivery time refers to the time period between the ordering of products and their availability to the customer [12], typically measured in working days [13]. Since this definition does not describe the process in between those steps, Christopher [14] expands it to a more comprehensive view of the supplier, which includes the entire value creation cycle from the order placed by the customer, through the order processing and the transport to the receipt of delivery by the customer. This definition also serves as a reference for this paper and is divided into the individual, successive steps of the multi-level order-to-delivery process in a practical application. Furthermore, the extraction of relevant data for delivery delay prediction in each process step is a possibility. Examples of such data are listed in brackets in the following. The process includes receiving the order and confirming it with an estimated delivery date (estimated delivery time) [15], integrating it into the system and planning production (material data) [16],

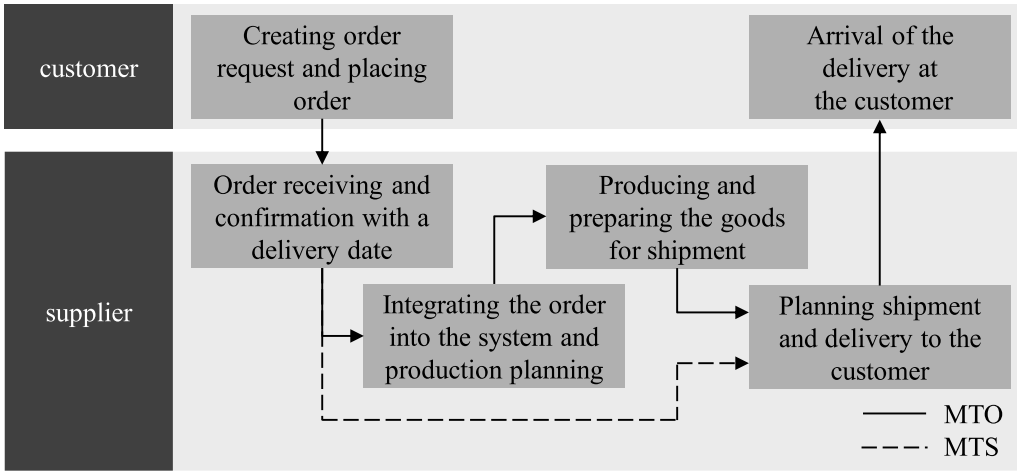

Fig. 2. Multi-level order-to-delivery process in production industry.

producing and preparing the goods for shipment (logistics data) [17], planning shipment and delivery to the customer (revised delivery date) [18], and finally, the arrival of the delivery at the customer which ends the whole delivery process (actual delivery date) [19]. The following Fig. 2 shows the order-to-delivery process through a flow diagram. Here, a further distinction must be made, that pre-produced goods can also be sold directly from stock like in the make-to-stock (MTS) process, which means that in contrast to the make-to-order (MTO) process production planning and production are not relevant for the delivery time which is therefore faster [20].

This shows that the SCM process from order to delivery involves a lot of individual processes, so that the delivery time and thus the quoted delivery date is influenced by various factors. According to Wiendahl [21] delivery date is distinguishing between the planned, requested and confirmed delivery date: The planned delivery date is set internally by the supplier, the requested date reflects the customer's request, and the confirmed date is the binding agreed-upon time following resource reconciliation. This subdivision is important for understanding the various prediction parameters later in this paper and is therefore explicitly mentioned here.

The timely fulfillment of delivery dates is an important and fundamental goal in SCM [22] as delays represent a significant challenge and can affect the entire supply chain [23]. It is not only important from the suppliers point of view, but also crucial from the recipients point of view to ensure that production processes run on schedule and, as a result, that customer orders are completed on time [24]. To define delivery delay the Association of German Engineers [25] describes delivery delay as the period between the delivery date agreed upon and confirmed by the customer and supplier and the actual delivery date. A positive value indicates a late delivery, while a negative value indicates an early delivery [25]. Given the central role that meeting delivery dates plays in the successful completion of production and assembly processes [26], the prevention and management of delivery delays is a key aspect [23]. As a result, the ability to predict potential delivery delays is becoming increasingly important to take proactive measures and avoid failures in the supply chain [27].

In the past, delivery delays were therefore either predicted on the experience of longstanding employees [28] or quantitative modeling approaches based on stochastic methods were applied [29]. However, manual estimation in particular proves to be particularly sensitive to errors due to the strong humanistic focus [30]. With the increasing advancement of predictive data analytics, prediction is now one of the most important modern tools in this context [31,32] that relies on the use of AI and big data analytics (BDA) to process supply chain data in SCM. A key application of predictive data analytics involves ML-based forecasting models that utilize historical data and external influencing factors to predict potential delays in the procurement process enabling potential countermeasures to be derived at an early stage [6]. To obtain a comprehensive overview of the current state of the art regarding BDA and AI in SCM, it is essential to examine the extent to which the literature addresses the topic and whether the focus is on the area of predicting delivery delays using predictive algorithms. Several systematic literature reviews have already examined the use of BDA and AI in SCM. However, it should be highlighted that most publications look at the topics more generally and focus less on individual applications like the prediction of delivery delays.

2.1. Big data analytics in supply chain management

In the area of BDA in SCM, Nguyen et al. [33] provide a comprehensive and overarching overview of BDA applications in SCM focusing on decision optimization, efficiency improvement and risk reducing. Furthermore, Wang et al. [34] specifically analyze the use of BDA in logistics with a focus on operational application and transport optimization to increase efficiency. Similarly, Maheshwari et al. [35], address the topic of BDA applications in SCM and focus on risk analysis and demand forecasting in global supply chains. Nevertheless, despite the thematic connections between these SLRs on BDA in SCM and logistics and the thematic links between the topic of delivery delays and their prediction, the topic remains largely unexamined in the extant publications.

2.2. Artificial intelligence in supply chain management

A review of the extant literature on the subject of AI in SCM reveals a gap in the prediction of delivery delays as well. For instance, Toorajipour et al. [36] provide an overview of current and future applications of various AI models in SCM focusing on topics like demand forecasting, risk management or process management. Similar, Baryannis et al. [5], focus on the application of various AI models in Supply Chain Risk Management and particularly highlight the use of predictive algorithms. Only Brintrup [37], who classifies and examines AI applications in SCM, particularly regarding the use of predictive models, explicitly mentions the topic of delivery delay prediction. However, only a single example is mentioned without any associated analysis or further explanation.

2.3. Forecasting in supply chain management

In addition to these broader literature reviews on BDA and AI in SCM, some publications address more specific topics such as setting delivery dates and general forecasting in the supply chain. For example,

	Characteristics	Categories			
1	Focus	Research Outcomes	Research Methods	Theories	Practices or Applications
2	Goal	Integration	Criticism	Identification of Central Issues	
3	Perspective	Neutral Representation	Espousal of Position		
4	Coverage	Exhaustive	Exhaustive with Selective Citation	Re-presentative	Central or Pivotal
5	Organization	Historical	Conceptual	Methodological	
6	Audience	Specialized Scholars	General Scholars	Practitioners or Policy Makers	General Public

Fig. 3. Categories relevant to this systematic literature review according to Cooper's methodology [50]. Relevant categories highlighted in grey.

Bhalla et al. [38] provide a comprehensive overview of tools and methods for the estimation of delivery dates in an engineering-to-order environment, focusing on mathematical models for lead time estimation and capacity planning. Furthermore, Syntetos et al. [39], examine the gap between theory and practice in the use of forecasting models within supply chains focusing on improving planning accuracy. Although these publications aim to calculate delivery dates and focus on forecasting models in supply chain contexts, the publications neither address the actual prediction of delivery dates or delays.

Concluding, several literature reviews have already examined the application of BDA and AI in SCM and the use of predictive algorithms in supply chains. Despite the extensive search, however, to the best of the authors' knowledge, no studies could be identified that comprehensively review and summarize existing models for the prediction of delivery delays and fully elaborate on the current state of research. As a result, the topic also lacks a summary of future research needs that clearly presents all open research questions and organizes them along an agenda. Given the constantly increasing challenges for supply chains and adherence to delivery dates, as well as the existing possibilities for predicting delivery delays, this represents a deficit in the existing literature. Since the use of models to predict delivery delays and the resulting information about late deliveries can offer considerable potential for savings along the entire value chain in a company, the availability of information in the relevant subject area has to be improved. For this reason, there is an urgent need to synthesize and evaluate the current state of the art in delivery delay prediction and, on this basis, to develop a research agenda for the further development of the research area.

3. Research methodology

The present paper employs a SLR methodology to identify and analyze the state of the art regarding the prediction of delivery delays and to derive an agenda for future research. The SLR method is particularly well-suited to answer research questions that emerge from the synthesis of results from multiple studies that address similar issue [40, 41]. This involves identifying publications relevant to the topic area by applying systematic methods [42] and summarizing the qualitative and quantitative results [43]. The SLR method ensures transparency across all review steps and effectively identifies future research needs through analysis and synthesis of existing studies [44,45]. This also clarifies the decision to conduct an SLR rather than alternative methods, such as a meta-analysis or a scoping review. A meta-analysis is a statistical technique that aggregates results from multiple studies, whereas a scoping review, by its nature, is generally less comprehensive. Furthermore, Tranfield et al. [46] emphasize that an SLR not only advances theoretical development but also facilitates practice-oriented progress by systematically aggregating knowledge from numerous publications which is a clear objective of this SLR. Thereby, the current state of the art should be summarized completely, comprehensively, transparently and reproducibly [47]. This paper is based on the five steps of the procedure model of vom Brocke et al. [48], as this is a methodologically well-established and widely accepted approach in review theory [49]. It begins with the definition of the scope of the systematic literature review (Section 3.1), followed by the conceptualization of the topic (Section 3.2). The subsequent literature search (Section 3.3) is followed by the literature analysis and synthesis (Section 4). The final step is the derivation of the research agenda (Section 5).

3.1. Scope of the literature review

The scope of the literature review was defined based on the taxonomy for literature reviews developed by Cooper [50] (see Fig. 3). The research review focuses on the categories "research outcomes", "research methods" and "practices or applications" with the specific aim of highlighting central problems in research and critically challenging the results. Therefore, this review considers the entirety of the literature from a neutral perspective, but only cites the relevant sources, which means that the review is categorized as exhaustive and selective. The review is structured based on a conceptual structure and addresses the target group of specialized scientists.

The review of the existing literature provides a well-founded overview of the applications for predicting delivery delays. The aim of this systematic literature review is therefore to summarize the current state of the art of prediction of delivery delays, including the algorithms and data used. It also aims to determine to what extent the models used are already being applied in practice and how extensive their general practical applicability is. Since this is not adequate for comprehensive consideration of the topic, the aim is also to determine the open research questions in the current state of research and to derive an agenda for

Table 1
Detailed research questions and corresponding methods.

Label	Research question	Corresponding method
RQ1	What prediction approaches are currently used to predict delivery delays and how do they affect model quality and prediction accuracy?	Systematic literature review/ framework analysis
RQ2	Which data sources are used for the prediction of delivery delays and how do they impact the model quality?	Systematic literature review/ framework analysis
RQ3	Are predictions of delivery delays validated on a theoretical level through simulations or through use cases based on real data sets?	Systematic literature review/ framework analysis
RQ4	What implications for further research questions are elaborated in the current literature and what are corresponding potential approaches?	Systematic literature review/ development of the research agenda

further research with a particular focus on the identified research gaps. To achieve these aims, the general added value of a research agenda for further research must first be explained. The creation of a current and structured research agenda is therefore highly relevant, as it is necessary from an efficiency point of view to conduct research in a target-oriented approach along a clearly defined agenda. Such an agenda makes it possible to structure further research topics transparently and to effectively bundle them. As there is no presentation of the current state of research in delivery delay prediction, there is, to the best of the authors' knowledge, no agenda for future research. However, a clear presentation of open questions and identified deficits is essential to contribute specifically to closing the present research gap. To differentiate the topic from other forecast approaches in the supply chain and to keep the research agenda stringent in terms of scope, the review process focuses only on the prediction of delivery dates and the resulting delays. Approaches that deal with the forecasting of lead times, the breakdown of the supply chain or risk management in the supply chain are intentionally disregarded due to the lack of traceability for a specific delivery delay. This distinction ensures that the research agenda is thematically consistent and that the scientific focus is clearly defined. In contrast to the prediction of delivery delays, it is also not certain that these approaches include all factors that affect the delivery delay in the prediction. In addition, the use of many well-established ML approaches and extensive data collection along the supply chain offers considerable potential for directly predicting delivery delays based on historical data. This results in the following four research questions (see Table 1) addressing the research gap. The table also shows the methodological steps required to answer each research question.

3.2. Conception of the review topic

Before a review of a topic can be conducted and the state of the art can be summarized, a knowledge base must be established according to [51] to generate synthesized knowledge of the subject area and to be able to conduct the review adequately. Based on the explanations given in Sections 1 and 2 and a review of 65 exploratively researched publications, the most relevant keywords for the topic covered in this paper were identified and contextualized. For this purpose, the relevant keywords were systematically identified and documented in tabular form, with duplicate occurrences aggregated accordingly. Subsequently, the relevance of these keywords was assessed collaboratively by all authors and two additional experts in the field of SCM. In addition to the explorative research, all identified key words were researched in the IEEE Xplore database, individually and in combinations, to ensure thematic relevance. The papers found were analyzed for thematic relevance to the objective of this systematic literature review. In this way, it can be ensured that the key words used in the research cover a broad spectrum of the terms used in the literature. The result of this preliminary work is a concept map [52] for delivery date prediction which is shown in Fig. 4. The concept map includes all synonyms that were not excluded by the previous content check and are integrated into the search string as a starting point for the SLR.

3.3. Literature search

Based on the concept map, the key words were expanded using wildcards and transferred to the following search string using Boolean operators: ("predict*" OR "forecast*" OR "anticipat*" OR "estimat*") AND ("procurement" OR "supply chain" OR "deliver*" OR "purchas*" OR "buy*") AND ("on time" OR "delay*" OR "lead time" OR "timely"). The use of AND operators guarantees that the thematic relevance is increased by excluding publications that only focus on one area of the subject field. The search strategy was extended by the approach of the STARLITE framework [53]: This SLR is limited to English-language journal articles and conference papers published between 1970 and 2024, which were identified through searches in the electronic databases Science Direct, Web of Science, EBSCOhost, IEEE Xplore, JSTOR, Springer Link, ACM Digital Library, Taylor and Francis, Dimensions AI and BASE. The exclusive selection of journal articles and conference papers is intended to ensure the scientific quality of the publications considered for this review through the peer review process, which is required for these types of publication.

By applying the search string to the metadata title, abstract and keywords using the STARLITE methodology, a total of 342,774 publications were extracted from the databases listed. To identify the relevant publications, the PRISMA flow diagram by Moher et al. [54] was used to identify the relevant publications (see Fig. 5). According to the procedure, first all duplicates, and incorrect data sets (e.g. incorrect language) were removed, leading to a new number of 184,781 potentially relevant publications. In these remaining 184,781 publications, various papers were diagnosed that did not correspond to the topic of delivery delay prediction, as individual databases incorrectly applied the search

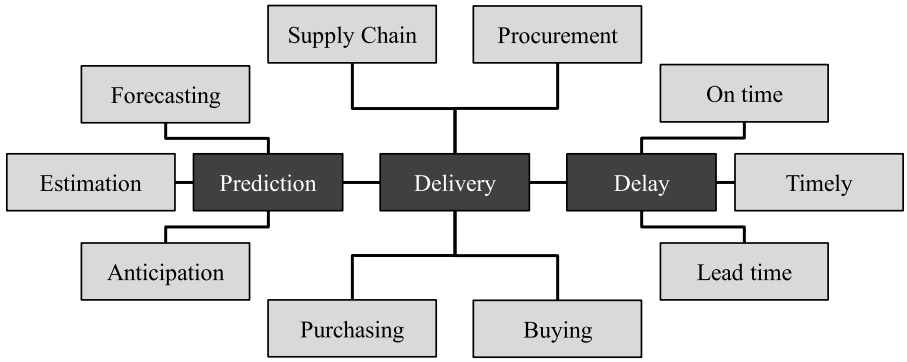

Fig. 4. Concept map for delivery delay prediction according to Rowley & Slack [52].

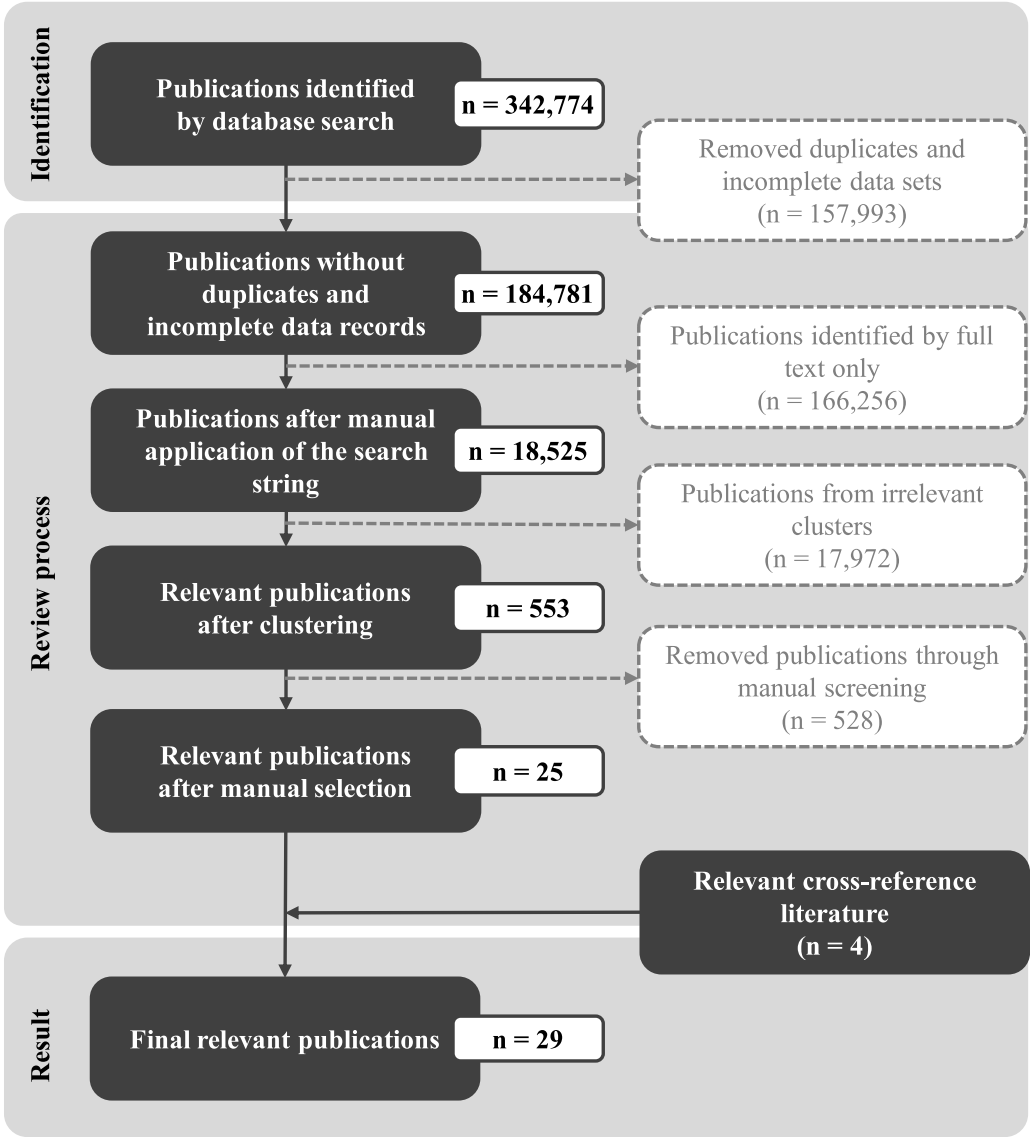

Fig. 5. Literature selection process according to the Prisma scheme with the number of publications per selection step by Moher et al. [54].

Table 2
Quality criteria for literature screening and evaluation.

QC1	Publications addressing the area of supply chain/supply chain management are included.
QC2	Publications addressing the area of delivery dates in the procurement process are included.
QC3	Publications addressing the area of forecasting, estimation, anticipation or prediction using machine learning are included.
QC4	Publications addressing the logistics of manufactured goods are included.

string to the full text of the publications. After manually applying the search string to the title, abstract and keywords, a number of 18,525 relevant publications remained for the evaluation phase.

In the next step a literature screening is carried out and subsequently the publications are evaluated based on the full text. The following four quality criteria (QC) (see Table 2) were defined for the screening and evaluation process:

The screening of the publications was based on the Natural Language Processing clustering approach according to Weißer et al. [55]. In preparation for the application of k-means clustering, word separation and removal of all stop words were carried out. By clustering, the most relevant (most frequently occurring words) per cluster are identified to determine the thematic orientation of the cluster. Since the clustering was carried out based on title, abstract and keywords without the corresponding terms of the search string, a dimensionality reduction was carried out using latent semantic analysis to reduce the large text corpus and improve the clustering results according to Refs. [56] and [57]. To further improve the clustering results and fulfil the established quality criteria, the analysis is not carried out solely based on the top words, in contrast to Weißer et al. [55]. For further validation, a representative and statistically validated sample size of each cluster was calculated based on the assumption of homogeneous clusters and in accordance with the findings and mathematical approach described in the publication by Israel [58]. The respective publications of this sample were reviewed for compliance with the QC based on the abstracts. If at least one publication in the sample meets the QC, the entire cluster is classified as relevant. Therefore, it can be ensured that no relevant publications are excluded by the clustering process and that the corresponding cluster is only classified as irrelevant if the top words do not match the scope and the QC 1–4 are not fulfilled by the publications in the representative sample.

To further identify the relevant publications, the remaining 18,525 publications were processed by the cluster algorithm in preparation for

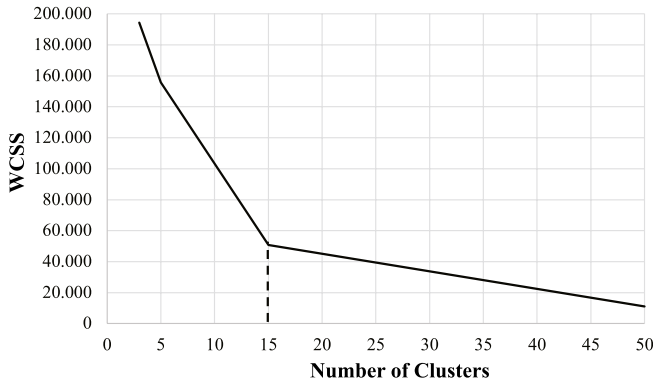

Fig. 6. Elbow curve resulting from the clustering process indicating the optimal number of clusters at k = 15.

Table 3
Clustering table with top words, cluster size and assessed relevance.

Cluster number	Top words	Cluster size	Relevance
1	Network, packet, communication, wireless, data	2855	No
2	Alcohol, drink, discount, behavior, intervention	59	No
3	Patient, cancer, treatment, therapy, care	1198	No
4	Abstract, may, email, copy, user	129	Yes
5	Drug, delay, kidney, donor, vaccine	3543	No
6	Route, network, node, protocol, mobile	2072	No
7	Project, construct, cost, software, manager	947	No
8	Reward, neuron, task, rat, cortex	611	No
9	Cache, content, network, user, edq	1137	No
10	Video, stream, network, quality, quote	980	No
11	Pregnancy, birth, women, infant, preterm	1588	No
12	Demand, inventory, product, forecast, bullwhip	191	Yes
13	Breastfeed, mother, initial, feed, infant	119	No
14	Model, system, data, predict, product	233	Yes
15	Care, health, service, patient, data	2863	No

the evaluation phase. The ideal number of clusters was determined by applying the elbow method. When applying the elbow method, the optimal number of clusters corresponds to the point at which the plotted curve flattens significantly, indicating a distinct bend (the "elbow"). In this analysis, this inflection point occurred at 15 clusters, thus determining the optimal cluster count as 15 (see Fig. 6).

The corresponding top words, cluster size and relevance assessment are shown in Table 3. After analyzing the top words and the representative sample, clusters 4, 12 and 14 with a total number of 553 publications were defined as relevant. It is evident in cluster 4 that the top words do not align with the subject of the SLR. A subsequent analysis of the representative sample revealed that this cluster contained a relevant publication which may be explained by the mention of the top word “abstract”.

Subsequently, all abstracts of these remaining publications were analyzed regarding the defined QC 1–4. After manually excluding the irrelevant publications that were thematically inappropriate or did not meet the quality criteria, the resulting publications were analyzed based on the full text, resulting in a final number of 25 publications. To exclude errors and eliminate subjective biases throughout the literature selection process – both during abstract screening and full-text analysis – each step was independently performed by all authors. The comparability of the initial assessment was evaluated using Cohen’s kappa analysis according to Cohen [59]. An agreement of 0.96 was achieved, which according to Landis & Koch [60] corresponds to an almost perfect result. The uncertain publications were discussed collectively, so that publications were excluded only if all authors agreed on their irrelevance for the underlying research questions.

It can be noted that the necessary combination of all QC was not guaranteed in many of the publications relevant after clustering, which accordingly led to the exclusion of these publications. For example, some publications were excluded that took supply chain and delivery times into account but did not predict delivery times or delays (e.g. [61] or [62]). Other publications were excluded that did a forecast but not for delivery delays (e.g. [63] or [64] did a lead time forecasting). Similarly, publications predicting assembly start delays and internal logistics delays by workstation were also excluded e.g. [65]. As a result, the prediction criterion (QC3) and the delivery dates in procurement criterion (QC2) were responsible for many exclusions. As an additional safeguard, a forward and backward search was performed on the 25 publications to identify publications that were not identified by the initial application of the search string [66]. This procedure identified a further 4 publications, resulting in a final dataset of 29 publications. In this process, it was explicitly verified that the 4 additional publications were not excluded at any of the individual steps of the literature selection process. These publications were not included in the initial overview of publications because the criteria to which the search string was applied were not selected suitably in the four publications. However, given that these 4 publications were the only ones identified as relevant in the forward and backward searches of the relevant publications, it can be ensured that no relevant publication was excluded or overlooked in the process of literature selection. The relevant 29 publications are analyzed in the further approach according to vom Brocke et al. [48] in Section 4.

4. Results

The aim of this section is to bring the identified prediction models into an organized structure and to evaluate their content. In particular, the contribution of the literature to the research questions defined in this paper is highlighted and the starting point for future research is consolidated. Before analyzing the content, Section 4.1 develops a framework for the publications that will be used to classify them.

4.1. Development of the framework

The development of a framework is a common approach for structuring literature [67–69]. After analyzing the relevant publications, it has been found that the framework for answering the research questions addressed in the analysis includes five categories (see Fig. 7).

4.1.1. Forecast area

The first distinction in the relevant publications is the *forecast area*. According to Rokoss et al. [70] and Schéle et al. [71], a distinction can be made here between two primary objectives, as to whether the delivery date or the delivery delay is predicted. In case of predicting the delivery date the difference between the predicted and the confirmed date indicates the delay, which makes this distinction relevant. Therefore, the *forecast area* is divided into *delivery date prediction* and *delivery delay prediction*. The second level of differentiation distinguishes the industry sector from which the prediction originates. On the one hand, predictions are made based on information from the manufacturing company, see Rokoss et al. [70], on the other hand, predictions are also made based on data from an external logistic service provider, see Ghaithan et al. [72]. As these two categories partly use different data for forecasting, the framework differentiation *production industry* and *logistics* is made.

4.1.2. Machine learning model

Predicting delivery delays is a data-driven problem that relies on analyzing historical data sets, for example from ERP-systems. It aims to

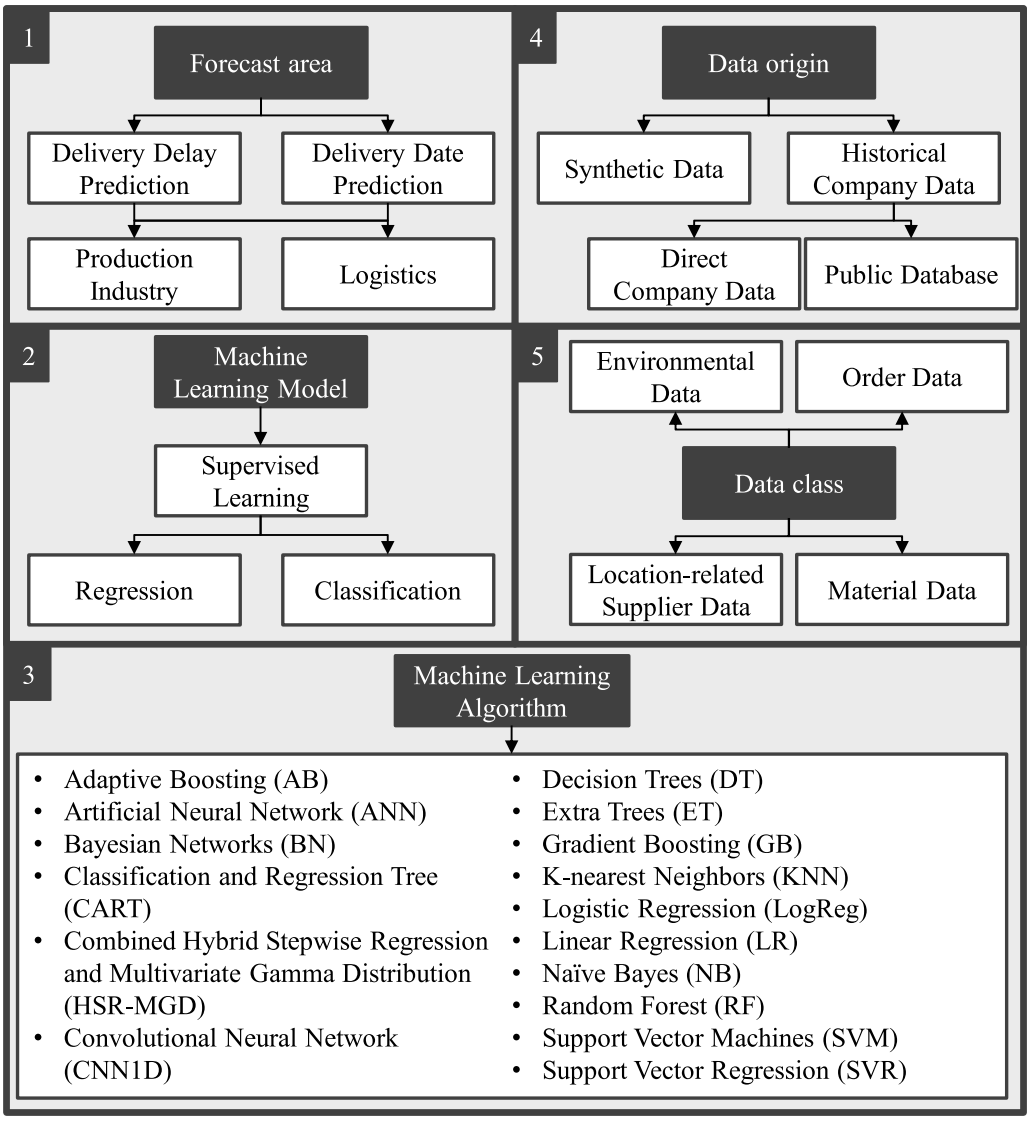

Fig. 7. Dimensions of the developed framework showing all categories and subcategories.

determine whether a delivery will be on time or to predict the duration of a potential delay. In this context, *supervised learning* is particularly suitable for predicting delivery delays as it relies on labeled datasets where the model learns from historical data to make future predictions [73,74]. This approach is particularly effective in this context as it enables the identification of patterns and relationships between input features (e.g. order details or supplier information) and the target outcome (e.g. delivery delay) [75]. In supervised learning, the distinction between classification and regression is crucial [76]. *Classification* is used when the prediction involves discrete labels or categories, such as predicting whether a delivery will be late or on time, effectively categorizing the results into defined classes [76,77]. *Regression* is used when the prediction involves continuous variables, e.g. estimating the exact duration of the delivery delay in days [76,77]. The choice between classification and regression depends on the type of outcome variable being predicted [76]. This clear description ensures that the chosen ML-model is tailored to the specific prediction task, which increases the accuracy and relevance of the results [77]. Reinforcement learning is aimed at dynamic scenarios in which an agent interacts with an environment and learns to make optimal decisions through feedback in the form of rewards or punishments [78,79]. Therefore, it is not suitable for predicting delivery delays. However, predicting delivery delays is a static problem that aims to make one-time predictions without an agent actively interacting with an environment or optimizing decisions. There is no environment that responds to actions, nor is there a mechanism for modeling dynamic interactions. Unsupervised learning is also unsuitable for predicting delivery delays. This method does not require a target variable and aims to identify patterns or clusters in the data [80]. However, for predicting delivery delays, there is a clear target variable such as occurrence or duration of delay, which means that only supervised learning can be applied, as confirmed by the relevant literature.

4.1.3. Machine learning algorithm

Predicting various delivery delays using ML algorithms [5,81] takes advantage of the ability of these methods to uncover complex patterns in large datasets and make accurate predictions [82]. In the field of ML, several algorithms have proven to be particularly suitable for this task. Therefore, the following algorithms are used in the relevant publications: *decision trees (DT), random forests (RF), logistic regression (LogReg), support vector machines (SVM), artificial neural networks (ANN), naïve bayes (NB), linear regression (LR), gradient boosting (GB), k-nearest neighbors (KNN), bayesian networks (BN), classification and regression trees (CART), convolutional neural network (CNN1D), extra trees (ET), adaptive boosting (AB)* and *combined hybrid stepwise regression and multivariate gamma distribution (HSR-MGD)*. The selection of the most suitable algorithm is predominantly determined by the requirements and the

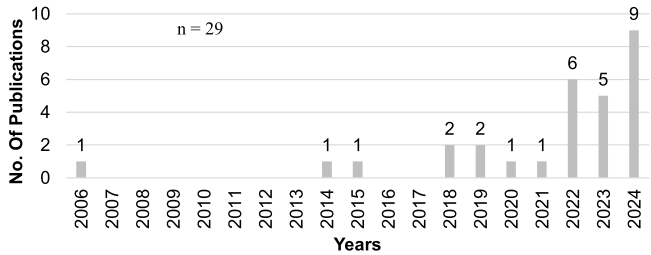

Fig. 8. Chronological development of the relevant publications.

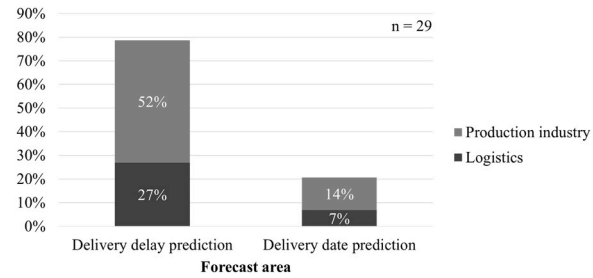

Fig. 9. Percentage distribution of the relevant publications regarding the forecast area.

characteristics of the data. Consequently, a range of algorithms is subjected to experimental testing in the relevant publications. It is important to acknowledge that certain hybrid techniques are grouped together, as evidenced by the categorization of XGBoost under GB. This grouping may potentially lead to a degree of ambiguity or overlap in the techniques' definition.

4.1.4. Data origin

A close examination of the relevant publications, considering the inductive formation of categories, shows that the direct prediction of delivery delays is fundamentally based on two different data origins. For example, Steinberg et al. [27] uses a historical data extract from a real company, whereas Bouhaid et al. [83] generates artificial data through a simulation. Resulting, the data origin in this paper is divided into the two categories of *historical company data* and *synthetic data*. However, the review of the relevant publications also shows that the historical data has another level of differentiation. Some studies use data sets that were provided directly in collaboration with a company [27,70], while others use data sets that were provided by an anonymized company from a public database [29,84]. As a result, the subcategories *direct company data* and *public database* are added.

4.1.5. Data class

A central distinction between the prediction models in the identified publications is the data class, which categorizes the corresponding category of the database used for the prediction. The first data class relevant for the prediction of delivery date delays is *order data*, which includes all specific data on article type and quantity, delivery times, quantities of the respective orders, etc. [27,70,85]. A second relevant data class is the *material data*, which relates to specific product properties of the end product to be manufactured, such as material composition, dimensions or weight, and is also of significant relevance for the prediction [26,28]. To include a geographical reference in the prediction, *location-related supplier data* is also used in some publications [72, 86]. This provides insights into location-specific data for each supplier, such as the country of origin, the region and the distance from the respective supplier. Since this information is also essential for the choice of transport mediums, this data class as well contains information on the selected delivery methods, such as road, air or sea freight. Last data class of the framework tis the *environmental data, which* includes all additional data on possible environmental influences during delivery, such as weather conditions, road closures or seasons [83,86]. This category of data is particularly relevant for the precise assessment and prediction of delivery risks in the supply chain [6].

4.2. Analysis

Based on the established framework, the 29 relevant publications were classified, and a descriptive and contextual analysis was conducted. The objective was to identify the state of the art of the models used for the direct prediction of delivery delays and the resulting prediction accuracy (cf. RQ1) and to determine which data sources are used for the prediction and how these affect the prediction accuracy (cf. RQ2). Furthermore, the practical applicability of the models was analyzed to ascertain whether the predictions are theoretically or practically validated (cf. RQ3). In addition, further research identified in the publications was examined and based on these findings, a research agenda was developed, which outlines the open topics for future research (cf. RQ4).

A comprehensive overview of the development of a research field can often be obtained by organizing the relevant publications chronologically (see Fig. 8). This reveals that the prediction of delivery delays begins in 2006, with only one publication addressing the topic. Afterward, there is a notable gap of several years, with interest in the subject gaining momentum from 2018 onwards. Since then, there has been a steady increase in publications, indicating a growing trend and highlighting the rising relevance of research focused on the direct prediction of delivery delays.

In the following, the individual categories of the framework are analyzed. In addition to the evaluation of the corresponding categories, content-related aspects of the individual publications are also addressed. Likewise, the need for further research mentioned in the publications is processed and analyzed.

4.2.1. Forecast area

The distribution of the 29 publications analyzed shows that the area of predicting delivery delays is particularly well represented. A total of 79 % of the studies deal with the prediction of delivery delays, while 21 % of the studies are dedicated to the prediction of delivery dates (see Fig. 9).

However, it is important to note that this subdivision was only made based on the problem description described in the respective publications. For example, Steinberg et al. [27] conducted a study on "late supplier deliveries", Thomas & Panicker [29] on "order delivery delay prediction", Zheng et al. [87] on "buyers predict order delays" and Hotta et al. [84] on prediction of "time delay in delivery". All these publications are clearly aimed at predicting delivery delays, which is why they are summarized in the category *delivery delay prediction*. In contrast, Benerjee et al. [26] examine the prediction of "supplier delivery times", but in their publication they determine the deviation of the delivery time in days from the delivery time confirmed by the supplier. Liu et al. [88] also address the prediction of "purchase orders delivery times" and then determine a prediction error in days compared to the supplier's predicted delivery time. A similar procedure can as well be seen in Rokoss et al. [70], who also carry out a "delivery time determination" and as a result determine the deviation of the delivery time from the forecast. Due to the lack of a direct reference, these publications are summarized under the category *delivery date prediction*. However, the description of the specific publications clearly shows that, despite different terminology of the forecast area, the result is always a prediction of the delivery delay. The prediction approaches used in the relevant publications are transferable in the areas of delivery delay prediction and delivery date prediction; similar data sets and models are used for the prediction. As a result, it can be concluded that different terms are used in the literature for the same forecast area and therefore there is no standardized definition or terminology of the problem. Therefore, this different way of characterizing the topic can be described as a misleading and incorrectly

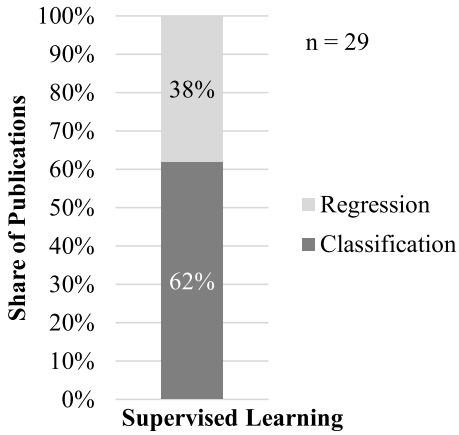

Fig. 10. Distribution of classification and regression in relevant publications.

created distinction.

Furthermore, a combination of the forecast area with the industrial sector (see Fig. 9) also shows that the percentage distribution of the individual industry sectors is almost the same for the two forecast areas. This confirms the finding of different problem designations for an almost identical prediction area, as none of the delay or date prediction terminology can be assigned to a specific industry sector. It is recommended for further research to adapt the terminology accordingly, so that both subject areas can be brought together and do not work on identical questions under different designations. Specifically, it is recommended to consolidate the research under the area of delivery delay prediction, as this topic focuses on the broader field of research, which includes both the delivery date and the delay prediction.

4.2.2. Machine learning model

The analysis of the ML models shows that, as stated in the framework's explanation, only supervised learning is used for the prediction of delivery delays. Looking at the distribution of these supervised learning algorithms, classification algorithms are used more frequently than regression algorithms, with a share of 62 % and 38 % respectively (see Fig. 10).

Based on the described fact that classification algorithms only make a binary decision (e.g. delay: yes/no), while regression algorithms predict the specific duration of the delay, there is already a deficit here due to the reduced scope and therefore added value of the publications using only classification. For example, Steinberg et al. [27] or Banerjee et al. [26] predict specific delay durations using regression algorithms, whereas Thomas and Panicker [29] or Baryannis et al. [6] only estimate the probability of a delay occurring, but without a delay duration. Even though the classification algorithms can only make a binary decision, classification offers the possibility to split the delay prediction into classes such as: 0–1 days delay, 1–2 days delay, etc. Although this is less accurate than an exact determination of the delay duration using regression algorithms, it offers increased added value compared to a simple yes/no statement about the delay. A closer look at the publications using classification algorithms shows that only Rezki [7] and Zaghdoudi [89] use such a classified subdivision of the delay duration for a more accurate prediction. The remaining publications, such as Wyrembek [81] or Hudnurkar [90], only determine the probability of the delay without a reference to a duration. In summary, these two findings of the lower percentage of regression algorithms used and the possibilities of classification algorithms that are often not used show a deficit in current research. The information on the duration of a delay is particularly relevant for industrial applications, as a delay of one day compared to one week can have a significant impact on production planning and subsequent processes and therefore on sales [91]. This is because delays in delivery can delay or even abruptly stop planned

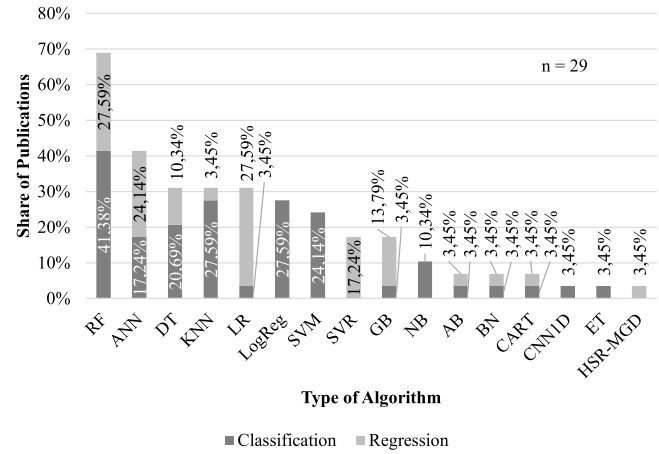

Fig. 11. Overview of the distribution regarding different algorithms used in the relevant publications.

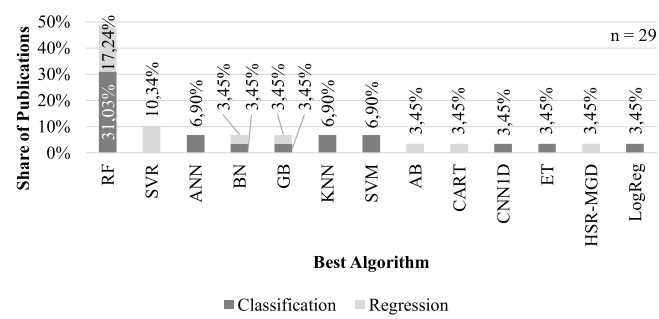

Fig. 12. Distribution of best performing algorithms in the relevant publications.

production. As a result, the company can no longer keep its own promised delivery dates, which can result in penalties and late payments. Therefore, an increase in the application of regression algorithms for specific prediction of the delay duration is recommended for future research. Likewise, the use of classification algorithms is recommended, considering the classification of delay durations. If the intervals of the classified deceleration are chosen correspondingly narrowly, added value can also be achieved in this manner. This possibility has already been demonstrated in other research areas, such as the prediction of flight delays. For instance, Xie et al. [92] and Rebollo & Balakrishnan [93] employ classification techniques to predict delays in air traffic, categorizing them into distinct time intervals.

4.2.3. Machine learning algorithm

A detailed analysis of the algorithms (see Fig. 11) used shows that RF are applied most frequently in both cases, with a share of 41 % for classification in studies like [94] and [95] and 28 % for regression in studies like [96] and [7].

Since RF is an ensemble learning approach, it can effectively handle both binary classification and numerical regression with high precision [97,98]. Besides RF, the most used algorithms for classification are LogReg with 28 % in studies such as [99] and [100], KNN with 28 % in studies like [101] and [102] and SVM with 24 % in studies like [86] and [72]. LogReg is particularly suitable for binary classification problems and offers good interpretability of the results [103]. KNN and SVM are also well-established models frequently used in classification tasks, especially when dealing with non-linear or complex data structures [104,105]. In the field of regression algorithms, alongside RF, the use of LR dominates with 28 % in studies like [86] and [102], ANN with 24 % in studies such as [96] and [70], and SVR with 17 %, as seen in [27] and

[72]. ANN and SVR provide models capable of modeling complex, non-linear relationships, making them highly suitable on large data sets with many features [106,107]. LR, however, is a simple, easy-to-interpret model that performs well for linear relationships between variables [108]. A detailed comparison showing the different algorithms used in each publication is shown in Table 6 in the appendix.

The additional analysis of the best performing algorithms per publication (see Fig. 12) shows that RF consistently delivers the best and most accurate predictions with a considerable majority of 48 % in both classification and regression algorithms.

Furthermore, SVR was also identified as the best algorithm in several publications dealing with regression. In the area of classification, ANN, KNN and SVM were the second most frequently identified best performing algorithms. The remaining best performing algorithms were each identified only once in the relevant publications as best performing. One explanation for this result can be linked to the lacking use of RF or SVR as a reference algorithm, as in Refs. [28] or [109]. This is partly because some papers only use or further develop one algorithm without including an additional algorithm for performance comparison, such as in Ref. [26].

Moreover, when assessing the best performing algorithms per paper, it is important to acknowledge the inherent limitations in statistical comparability of the results. This challenge is exemplified by the inability to adequately compare results based on metrics such as accuracy or root mean square error (RMSE). One possible reason for this limitation arises from the lack of comparability of the data sets. The data sets vary significantly in size, with some encompassing several hundred thousand data points and over 30 features, while others contain only a few tens of thousands of data points and fewer features. Given the distinct application of each prediction, the application would need to be standardized (e.g., the same number of features) to ensure comparability. However, this finding highlights the versatility of RF in providing optimal prediction results for a wide range of applications, as RF was identified as the best algorithm in 48 % of the applications in the identified publications.

In this context, enhancing the interpretability of prediction models is an important consideration. To transfer prediction models into real-world applications, the interpretability of the results must be increased, possibly by using techniques such as Shapley Additive Explanations (SHAP) or Local Interpretable Model-agnostic Explanations (LIME) [110,111]. These tools provide post-hoc explanations by attributing prediction outcomes to individual input features, allowing decision makers to understand and validate model behavior. In real-world supply chain applications, this transparency is essential to build trust in automated systems and to support user acceptance, particularly when integrating predictive outputs into ERP systems or decision support tools.

Combining the analysis results of the algorithms used and the respective best performing algorithms also addresses RQ1. Both classification and regression algorithms are used for predicting delivery delays, with a majority using classification algorithms. A closer examination reveals that RF is not only the most frequently used algorithm in both fields, classification and regression, but also consistently delivers the best prediction results. Although RF frequently achieves high levels of prediction accuracy due to its ensemble architecture based on multiple decision trees, RF models are comparatively more computationally intensive than simpler methods such as logistic regression [112]. Consequently, when selecting algorithms for practical applications, it is essential to consider not only predictive performance but also computational efficiency and scalability and thus the cost-benefit ratio.

In the field of regression, the SVR algorithm was second most frequently identified as the best-performing algorithm in the relevant publications. In the field of classification, the algorithms ANN, KNN and SVM were second most frequently identified as the best algorithm. It is noteworthy that the relevant publications rarely explain why one algorithm performs better than another. This inquiry is particularly interesting in cases where RF is not identified as the best algorithm. Apart from the fact that RF is generally not used in some of the publications that define other algorithms as best, no conclusive reason can be deduced here. This phenomenon has been documented by Rosenthal [113], who emphasizes this form of publication bias and demonstrates that studies yielding positive results are given greater prominence. In this case, literature could then highlight the best and most effective algorithms, while paying less attention to the poorer or less effective ones or even excludes them from publication. This results in an over-representation of successful algorithms, which may lead subsequent publications to overlook less successful results and reapply the same algorithms. To counteract the potential risk of publication bias, all identified publications, and all algorithms implemented in these publications were included in the analysis, regardless of the reported predictive performance. Furthermore, the frequency counts of algorithms were utilized descriptively rather than as quality indicators to avoid implicit weighting of successful approaches. However, it is crucial to acknowledge the absence of a systematic analysis of this potential bias, underscoring the necessity for further research in this domain.

It is also noteworthy that numerous publications continue to utilize and evaluate various algorithms that have already been demonstrated to be ineffective in previous studies. The repeated utilization of a limited set of algorithms can be ascribed to their demonstrated efficacy on structured data, their pervasive availability in machine learning libraries, their resilience in terms of usability, and the authors' objective to ensure the comparability of their results with those of other publications. Nevertheless, this convergence may also impose limitations on methodological innovation and potentially reinforce bias toward well-established techniques. To address this issue, it is crucial that future studies employ a more systematic justification process for selecting algorithms. This process should consider criteria such as data type, interpretability, computational cost, and application context. This approach will facilitate the further development of algorithms that have already demonstrated their usability and efficacy.

The distribution of algorithms raises the question of why classification algorithms are still predominantly used in research, even though regression algorithms, offer higher practical benefits for industrial applications. Likewise, the question arises why the full potential of classification algorithms for predicting the duration of delays is not used in most of the relevant publications. To optimize the practical benefits derived from this research, it would therefore be more advantageous to realign the research focus towards regression algorithms on the one hand and to extend the scope of application of classification models to include the classification of delay durations on the other. In this context it is important to distinguish the advantages of each model. While regression algorithms provide an accurate prediction of delay duration and are therefore well suited for detailed planning, they can be computationally complex [112]. Classification algorithms, however, can offer advantages in scenarios where interpretability, robustness, or computational efficiency are in the foreground [114]. Classification is often less sensitive to outliers and noise, and the division of delays into categories can simplify integration into rule-based (decision) systems or support clearer decision thresholds in operational contexts. A clear distinction and specific further development would significantly increase the usability of the prediction models for specific industrial applications.

Similarly, future research should better investigate the specific reasons for algorithm performance differences and highlight why and in which scenarios certain algorithms perform better than others. This could potentially reduce the number of tests in which models are retested that have been identified as less performant in the past. In addition to the further development of the prediction models, the actual implementation of these models in practical applications would be beneficial. As demonstrated by the relevant publications, the use of models for predicting delivery delays has proven to be successful. Instead of further optimizing prediction models in the single-digit range,

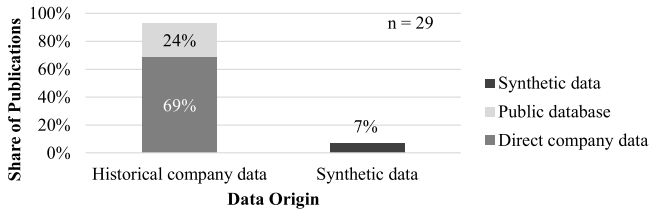

Fig. 13. Percentage distribution of data origin in the relevant publications.

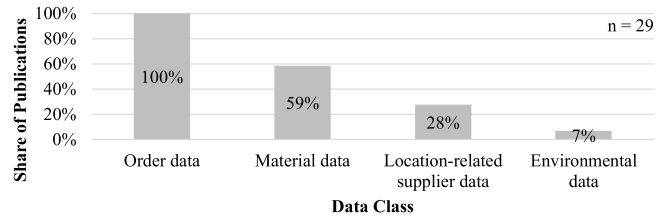

Fig. 14. Percentage distribution of data classes used in the relevant publications.

a cost-benefit analysis would be more appropriate here, particularly in the context of integrating these algorithms into industrial applications, e.g. through direct integration into ERP systems. Such an analysis shows the best possible and most efficient ways of integrating the prediction models into industrial applications.

In addition, as previously indicated, practical factors such as computational efficiency and deployment feasibility play a critical role in the selection of algorithms. These factors must therefore be considered as well. High-performing models, such as ensemble methods, frequently require substantial resources during training, including significant time requirements, considerable memory usage, and high computational load. For instance, RF and GB can require substantial processing time on large datasets, particularly during hyperparameter tuning. Conversely, more simplified models, such as SVR or linear models, have been shown to offer accelerated training and inference times, making them more conducive to real-time integration within operational systems. Future research should incorporate quantitative comparisons of computational cost metrics, such as training duration, memory usage, or required hardware, into the evaluation of algorithms. This approach would enable a more balanced evaluation that incorporates both predictive power and practical applicability in industrial settings.

4.2.4. Data origin

When analyzing the data origin, a significant imbalance between the use of historical company data and the use of synthetic data is immediately apparent (see Fig. 13). In fact, 93 % of the relevant publications use historical company data sets to predict delivery delays and only 7 % base their prediction on synthetic data.

Within the historical company data, 69 % use direct data from a company, as in Refs. [87] or [109], and in 24 % of cases data sets from public databases are used, as in Refs. [115] or [84]. These two distributions show the considerable practical relevance of the topic. In the context of synthetic data, the corresponding data sets are generated by a simulation as in Refs. [116] or [83]. The significant dominance of predictions based on historical company data confirms that research is actively being conducted in collaboration with the industry and that attempts are being made to achieve results with practical applications. These findings also provide the answer for RQ3. The prediction of delivery delays is validated with a considerable majority through the application of real historical company data use cases. Nevertheless, two deficits can be identified which should be addressed by future research. While synthetic data currently plays only a minor role in the reviewed publications, it offers specific advantages, particularly for training models on rare events that are underrepresented in real datasets. The implementation of simulated data sets enables the systematic modulation of delivery delay conditions, thereby enhancing the resilience of the model in scenarios that deviate from typical conditions. Future research should explore the potential of synthetic data to augment real data, employing techniques such as hybrid datasets or transfer learning methodologies. However, it is crucial to acknowledge that the designed structure of the data sets may result in reduced realism. Consequently, the application of real company data sets is expected to maintain its relevance in the ongoing enhancement of prediction models.

The second deficit arises from the predictions based on public databases. Since many of the databases offer options for modifying the data sets or uploading arbitrary data, and since there is a lack of verification instances, it is difficult to verify the data quality when using data from public databases. To guarantee a minimum standard of quality in the data source, it is recommended that direct company data be used. Furthermore, it is recommended that the data sets, both from companies and from databases, are evaluated for a minimum quality using specific methods for data quality assessment, such as dimension evaluation, outlier detection, or benchmarking against gold-standard data sets. This will also increase the comparability of the data sets between publications. Future research in this area would benefit from more explicit reporting and comparison of data quality measures in predictive modeling studies. One potential approach involves the establishment of a benchmark data set for future studies or alternatively, the formulation of suitable recommendations. These recommendations could address critical considerations such as the necessary size of the data set or the optimal number of features. The objective of this recommendation would be to ensure the comparability of findings across diverse studies.

4.2.5. Data class

An analysis of the data class shows that order data is used to predict delivery delays in all publications (see Fig. 14). This high value is due to the relevance of this data in terms of content, as information about delivery times or quantities and other order details are fundamental features for a precise prediction of delivery delays [117].

Steinberg et al. [27] and Zheng et al. [87] for example use the order quantity, requested and confirmed order and delivery dates as well as the supplier's name for the prediction of delivery dates. A similar approach is taken by Palomo et al. [109], who also use the order date and the guaranteed delivery time but use the order costs as a variable instead of the order quantity. In contrast, Abouloifa & Bahaj [118] exclusively uses order data for prediction but include a large number of features, including even specific information regarding partial deliveries or orders that are judged as non-compliant. Furthermore, the material data is used in 59 % of all publications for the prediction of delivery delays. This category is also an essential feature, as it contains information on component dimensions or delivery weights, for example, which can be essential for delivery capability. Pani et al. [28], for example, use the dimensions and quantity of containers delivered, while Liu et al. [88] integrate the material group and material number into the prediction, or Thomas & Panicker [29] who also use the quantity of units and additionally the weight as features for the model. It shows that material data is also a frequently used data class for prediction.

In contrast to the frequent use of the two previous data classes, environmental data, with a share of 7 % of the relevant publications, and location-based supplier data, with 28 %, are used significantly less. Regarding environmental data, for example Bouhadi et al. [83] use extensive data on road and weather conditions and Viellechner & Spinler [86] use seasonal and weather data as well as information on piracy risks on the delivery route. The analysis of the prediction shows that the accuracy is improved by integrating this data. Regarding location-based supplier data only Biazon de Oliveira et al. [102], Ghaithan et al. [72] and Pani et al. [28] use specific information about the distance of delivery for the prediction. Viellechner & Spinler [86]

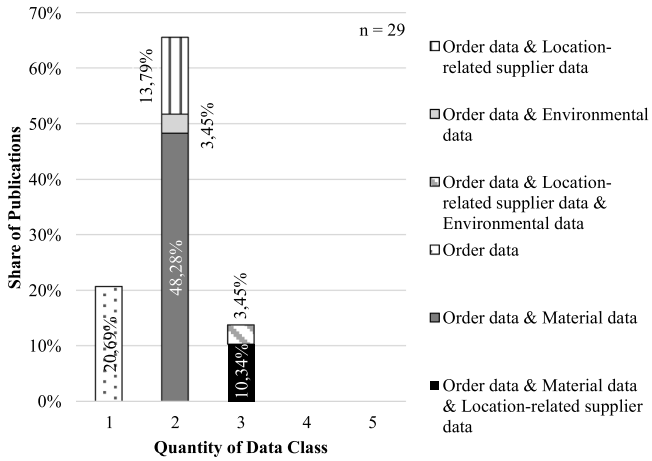

Fig. 15. Percentage distribution of combined data class used in the relevant publication.

modify this information slightly by referring to the distance as the time between two delivery ports as the "highest influence on shipping delay". Thomas & Panicker [99] and Camur et al. [96], in contrast, use information about the country-specific location without specifying the concrete distance of the suppliers for the prediction. In this context, it is noticeable that environmental and location data is used particularly in publications that carry out the prediction from the perspective of logistics.

A more detailed analysis of the data classes used for the prediction of delivery delays shows that 86 % of all identified publications combine two or fewer data classes in the prediction (see Fig. 15). When using a single data class, only order data is used, as in the case of Refs. [74] or [85]. All other data classes are not used individually, only in combinations. As soon as two or more data classes are used, order data is included in all combinations. With a share of 48 % of the relevant publications, the combination of order and material data is most frequently used here, as in the case of Refs. [6] or [100]. The combination of order and environmental data is used considerably less frequently with a share of 3 % as in Refs. [83]. The combination of order and location-related supplier data is also considerably rarer at 13 % like [83]. A combination of three data classes is only used by a minority of 14 % when combining order, environmental and location-related supplier data [86] or order, material and location-related supplier data [28]. Furthermore, no publication uses more than three data classes for forecasting.

Based on these findings, RQ2 can be answered. As can be seen, the prediction of delivery delays is always performed based on order data. The majority of publications also use a combination of several data classes, primarily order and material data, less frequently in combination with environmental or location-related supplier data. The analysis of the publications shows that the combination of data classes increases the prediction accuracy, as the features used are thus more varied and cover different aspects of the ordering and delivery process. Combining multiple data classes to predict delivery delays offers significant potential, which has identified three areas for further research. Future studies should investigate all combinations involving three of the data classes outlined in the framework, as well as the combination of all four data classes, to evaluate their predictive performance. For instance, further research could develop a model that considerably simplifies the overlay of ERP data with supplier location data or environmental data. One reason why this overlay is rarely performed despite better prediction results is the challenge of implementation. On the one hand, these data classes are structured differently and mapping environmental data to ERP data, for example, is not trivial. On the other hand, the technical challenge of using multiple data classes is increasingly larger. For example, cleaning the data set is more complicated, feature engineering is significantly more complex, and the models require more training time. Furthermore, external macroeconomic indicators, for example fuel prices, inflation rates, and financial indices at both the company-internal and country-specific levels, have the potential to considerably influence the performance of predictions. However, these factors are currently underrepresented in extant literature. Future research should explore the integration of such macro-level data into predictive models to capture systemic influences and enhance robustness, particularly in volatile global supply chain environments. Therefore, a comprehensive evaluation is necessary to assess their contribution to prediction accuracy in contrast to the company's specific data. This evaluation should be conducted in correlation with the existing data classes.

Nevertheless, the analysis of the relevant publications also shows that the publications do not provide a feature-specific analysis of the resulting improvement in prediction quality using the individual data classes, especially for location or environmental data. As already mentioned, it is only noted that the prediction performance is improved by including the other data classes. For future research, an analysis is recommended that explicitly examines the improvement in prediction quality for the combination of the different data classes. It is also important to ensure that data quality is as high as possible and that the various data classes can be combined despite the different types of data. To this end, a cost-benefit analysis is recommended for further research here. It should be analyzed how much improvement in the prediction is generated by the addition of the corresponding data classes and what costs are incurred in return by the procurement and processing of the additional data. It should also be considered whether the accuracy of the prediction increases enough to make the additional effort worthwhile. The analysis can also be extended to different industries, especially those with just-in-time versus just-in-sequence procurement.

4.2.6. Analysis of future research mentioned in the relevant publications

The analysis of future research needs in the field of predicting delivery delays identified in the relevant publications reveals some overlapping topics as well as individual findings that are crucial for advancing the field of research on the one hand and on the other hand offer considerable potential for providing specific added value for companies. Considering the overlaps in the identified open research questions, 16 of the relevant 29 publications highlight the importance of testing additional algorithms as an effective direction, making it the most frequently mentioned research need. Sarbas et al. [74], for instance, explicitly mention the need to explore further classification algorithms for prediction models in future research, whereas Thomas and Panicker [99] or Baryannis et al. [6] only recommend application of additional algorithms as a relevant future research focus. This indicates that some studies do not sufficiently emphasize the need for more precise research focusing on specific algorithms. Moreover, a clear tendency towards general statements without in-depth specification can be observed, which may lead to valuable resources being used on less differentiated and more redundant research approaches. The answer to RQ1 already shows that recent publications repeatedly test algorithms previously identified as less effective in older publications. Furthermore, most publications independently identify identical algorithms as best performing. In this context, it is noticeable that the reasons for performance differences are often barely mentioned or not mentioned at all, making it difficult to clearly assess the reapplication of these algorithms for future research. It is also evident that the use of classification algorithms is less effective in the context of prediction of delivery delays. This emphasizes the need, as stated in Section 4, to promote a more practice-oriented approach in future research by determining the actual duration of delays in each prediction. Therefore, fewer algorithms should be tested with a focus on single-digit optimization, specific reasons for performance differences should be identified, and greater emphasis should instead be placed on integrating proven algorithms into real industrial processes, for example through direct integration into

ERP systems. The proposed changes would result in substantial benefits for the industry, as the current models are already suitable for use.

Another research need, as indicated in the relevant literature, consists in the further development of the best-performing algorithm itself, which is described in 8 of the relevant 29 publications and is therefore the second most frequently described goal. For this purpose, various approaches are described that aim to improve the prediction accuracy. For instance, Pani et al. [28] refer to the integration of additional new features into the prediction model as the next step in their future outlook, while Thomas and Panicker [29] identify improving prediction accuracy by integrating additional features as a promising research direction. Baryannis et al. [6] also explicitly recommend the use of a dataset with more features in their future outlook. These improvement suggestions specifically address the selection and preparation of datasets. Further, analysis shows that 6/29 of relevant publications explicitly recommend improving prediction performance by including more data classes in the prediction. For example, Steinberg et al. [27] describe the inclusion of additional data classes such as the component description or the manufacturing process as beneficial, whereas Viellechner and Spinler [86] explicitly mention the use of more real-time data such as environmental data as necessary in further research. This aligns directly with RQ2 and the resulting further research which has shown that the analytical outcomes improves as the number of features and data classes increases. Future research should therefore focus on equipping models with more input features from different data classes. Specifically, models should be adapted to include real-time data such as weather or location data, which are not directly related to corresponding ERP data extracts, to optimize prediction results.

Table 4
Identified key gaps with corresponding further research needs.

Label	Identified key gap	Further research need/ recommendation
A	Lack of insights on how the number and combination of data classes impact prediction performance.	Conducting analyses to determine the effects of different data classes on the prediction performance to identify the optimal number and combination of different data classes.
B	Limited availability of regression models that incorporate real-time external data sources for predicting delivery delays.	Development of regression algorithms that can incorporate real-time data such as environmental data or location data into the prediction.
C	No cost-benefit analyses have yet been used to determine the optimal prediction timing based on the trade-off between early but less accurate and later but more precise forecasts.	Execution of cost-benefit analyses to determine the optimal time of prediction and to determine the most economical quality of the prediction models.
D	Missing practical approaches like frameworks or guidelines for integrating predictive models into ERP systems.	Integration of existing predictive algorithms into ERP systems to provide a direct benefit for the manufacturing industry.
E	Generalizability of prediction models across different industries has not yet been sufficiently investigated due to a lack of cross-sectoral validation studies.	Carrying out predictions in various industry sectors, especially in the differentiation between make-to-order and make-to-stock as well as corresponding multi-level supply chains to develop more robust models.
F	Presently, there is an absence of decision support systems capable of systematically translating prediction results into actionable operational recommendations for predicting delivery delays.	Development of decision support systems to generate automated recommendations based on the executed prediction.
G	The potential of multi-level or time-based classification schemes remains unexplored, with most studies using classification algorithms relying on binary delay prediction.	Use of classification algorithms to predict the duration of the delay by categorizing it into different durations, as compared to a simple yes/no prediction.
H	Federated learning has not yet been applied in scalable supply chain contexts, and implementation concepts for cross-company collaboration are missing.	Further development and application of the federated learning approach to utilize synergy and economies of scale in the general further development of predictive algorithms.

The analysis also shows that 7 of the relevant 29 publications recommend conducting additional use cases with real-world datasets to further validate and test predictive models. It is highlighted that use cases should be conducted in industries other than those previously investigated. The reason given for this is that extending the use cases to other industry sectors can improve the robustness of the models by using different features models. A broader application scope is essential for ensuring the transferability and adaptability of the prediction models. For example, Zheng et al. [87] propose testing models in various industrial sectors with different supply chain dynamics, while Steinberg et al. [27] suggest that it is necessary to simply apply the identified models in additional use cases for validation. These findings also align directly with RQ3 which has already shown that many publications apply models to directly predict delivery delays based on real company data. It is noteworthy that these predictions are largely based on practical use cases, and research continues to recommend this approach. Therefore, future research should also conduct further experiments with use cases from different industry sectors. It is recommended to distinguish between MTO and MTS production due to the usually different ordering processes. A distinction between less to multi-level supply chains, such as in the automotive industry, is also recommended due to the increased dynamics and the more complex dependencies.

In addition to the more general and frequently mentioned further research needs, the analysis of relevant papers identified three more specific research areas, each mentioned only in individual publications, but suggesting promising and forward-looking approaches. Pani et al. [28] for example, propose the development of a DSS which, based on prediction results, provides concrete recommendations for further action in addressing the results. Such a system offers considerable added value in managing delays and would allow less experienced employees to respond to delays with appropriate measures. This system can also benefit more experienced employees by providing them with early notification of delays, thereby empowering them to take proactive measures. There are already use cases which show that the application of a DSS for a different, but thematically related, problem has been successfully implemented in a pilot project [119]. As a result, this approach should be further explored in future publications. Another promising approach is mentioned by Liu et al. [88], who aim to optimize the prediction process by adjusting the training time so that even large data sets can be used for real-time prediction. To further improve utilization, a user interface is to be developed and tested that simplifies the application of the prediction models in daily operations. This approach simplifies the operation of the algorithm, thereby reducing the training period and the effective usage time. Consequently, this can result in time savings. This approach is closely related to the integration of prediction models into an ERP system and the introduction of a DSS. Moreover, Zheng et al. [87] propose an interesting approach by implementing a federated learning system that enables the anonymized use of data from different companies to predict delivery delays. This centralized platform could significantly accelerate research, by aggregating large datasets from various companies, including both customers and suppliers. At the same time, the system takes confidentiality aspects into account so that no sensitive internal company data becomes publicly accessible. This approach offers great potential for integrating prediction models more broadly and securely in practice, as it allows for a much wider and more straightforward application of predictive algorithms.

A review of the identified future research needs reveals a striking omission: the absence of any studies addressing the applicability and likelihood of adoption of predictive models in real-world practice. This is further confirmed by the results of the analysis in the previous

sections, which explicitly show that there is a need for further research in this area and that some practical research applications are easy to implement. However, it also shows that some of the future research needs address the applicability or adoption probability indirectly. For example, the fact that 7/29 publications recommend the use of additional real data sets from other industries shows that there is also a need for practical prediction models there. Similarly, the recommendation to develop and use a DSS or federated learning shows that some publications recognize and indirectly address the practical relevance of the prediction models. Nevertheless, the analysis shows that this important topic is clearly underrepresented and needs to be addressed specifically.

However, the analysis reveals that, despite growing interest in predictive approaches, key gaps remain in the literature, particularly regarding the practical implementation of predictive models, the systematic use and further development of regression models, and the further integration of multiple data sources such as environmental or location-related supplier data. Also, the individual publications do not clearly prioritize the further research needs mentioned, and therefore the next important research questions are not directly recognizable. It is also noticeable that many publications focus on the concrete further development of their specific application, meaning that overarching questions are not highlighted, and the research area is not fully considered as a result. The research area therefore lacks a clearly structured, goal-oriented agenda that specifically addresses clusters and prioritizes the open research questions.

5. Derivation of a research agenda for further research

The analysis of further research needs indicates that despite repeated mentions of several open research topics and the identification of interesting individual directions, the research field of prediction of delivery delays lacks a structured and goal-oriented foundation for future development. To achieve targeted progress and develop the topic efficiently, it is essential to address research topics in a coordinated and prioritized manner to avoid redundant efforts and focus resources on key questions. Based on the analysis conducted, a research agenda can be derived to structure future research topics to the best knowledge of the authors of this publication. In total, the following eight key gaps with corresponding further research need (see Table 4) were identified from the analyses in Section 4 that are relevant to the research agenda. The corresponding key gaps and further research needs are labelled with A-H to make the pairwise comparison clearer. The labels were assigned randomly:

To transfer the future research needs into a research agenda, it is necessary not only to derive and explain the content of the individual topics, but also to prioritize them.

To determine the prioritization of the research agenda in a systematic manner, the pairwise comparison method was employed. According to this approach, each identified future research need was individually compared with each other to establish a clear priority ranking. The outcomes of these comparisons were then visualized in a pairwise comparison matrix. Within this matrix, the future research needs (labelled A-H) were individually plotted and compared on both axes. If a given row's research need was considered more important than that in the corresponding column, it was assigned a value of 2. Conversely, if both needs were considered equally important, a value of 1 was assigned. Conversely, if the row need was considered less important than the column need, a value of 0 was assigned. Following the completion of all individual comparisons, the values within each row were summed, resulting in an overall priority score for each research need - the higher the sum, the higher the priority assigned.

To ensure practical relevance, the prioritization was conducted in close collaboration with industry experts. Specifically, 15 purchasing and production managers from German manufacturing companies participated in workshops aimed at evaluating the importance of each research need. The experts were divided into three groups of five

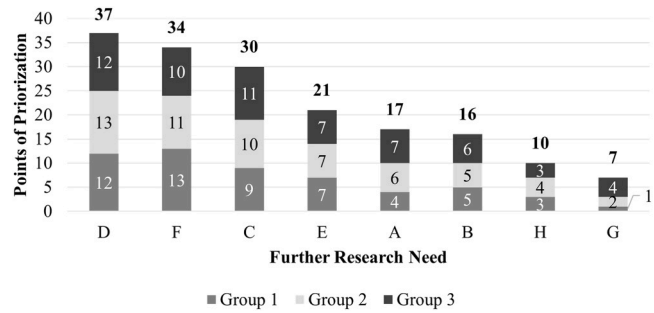

Fig. 16. Results of the pairwise comparison per group and cumulative result. The higher the overall bar, the higher the prioritization of the research need.

participants each. Each group independently performed the pairwise comparisons during dedicated workshops. The authors of this paper took on the role of moderators during these workshops, ensuring that the prioritization process reflected both practical and academic viewpoints. The aggregated prioritization results from each workshop group are summarized in Fig. 16 below using a stacked bar chart, which clearly illustrates the relative priorities assigned by each group. The detailed pairwise comparison matrices from each expert group are provided in Figs. 17–19 in the appendix.The results of the pairwise comparison led to a prioritization of D-F-C-E-A-B-H-G for the research agenda. This final prioritization was obtained by aggregating the individual results of the three groups into a cumulative score. This also ensured that discrepancies and different prioritizations between the three groups were considered. However, it should also be noted that the results of the three groups were consistently similar. To further justify the prioritization derived from the pairwise comparison, these eight research needs are summarized and explained again in the following.

A main aspect for generating practical value from research is the integration of existing predictive models into industrial applications (D). Achieving this requires a research agenda, that not only facilitates such integration but also identifies further improvements to enhance the performance of prediction models. In particular, the technical implementation within enterprise software environments of predictive algorithms could provide significant value by increasing efficiency and simplifying the applicability of predictive algorithms. By embedding predictive capabilities directly into existing company systems, these tools can be deployed quickly and effectively without the need for extensive modifications or additional infrastructure. However, it should be ensured that the models are easy to integrate and provide simple user guidance, enabling them to be used frequently. This approach could ensure that predictive models are accessible to decision-makers in real time, enabling them to optimize operations and improve decision-making processes with minimal effort. Therefore, this research need was selected as most important. It also creates a direct synergy with another area of essential importance for future research: the development of systems that provide actionable recommendations for managing predicted delivery delays (F). Integrating such a DSS into ERP systems would be highly beneficial, as it would create a user-friendly and easy-to-operate tool. This integration would not only simplify the management of delays, especially for less experienced employees, but also enable the automation of certain processes, reducing manual effort and minimizing the risk of errors. Furthermore, such systems could help decision-makers to evaluate multiple courses of action and choose the most effective strategy. While the integration of predictive models into ERP systems and the development of DSS offer significant practical benefits, their implementation also presents several challenges. These include technical barriers such as system compatibility and data availability, as well as organizational factors such as user acceptance, process adaptation, and compliance requirements. Similarly, the additional systems and their operation may result in additional costs. These challenges are further elaborated in the publication by Kaushik et al. [120],

which examines various obstacles regarding the integration of data analytics in small businesses. Another example of the emerging challenges is described by Kortian et al. [121], who examine a case study on the introduction of BDA analytics in a large railway organization and describe the arising challenges. The interpretability of the results must also be ensured and improved, using, for instance, SHAP or LIME. Future research should address how these models can be systematically applied to enhance the interpretability of high-performing models and increase trust in predictive systems within operational supply chain settings. Therefore, the potential barriers to ensure successful and sustainable implementation in industrial environments need to be analyzed.

In the context of integrating prediction models into ERP systems, another important consideration for future research arises: a cost-benefit analysis (C) to evaluate whether further optimizations in prediction models, particularly within the scope of marginal optimization, are truly justified. Such an analysis should assess the effort required to improve algorithms for minimal performance optimizations against the practical benefits of early predictions in the ordering process. Early-stage predictions, even when based on weaker data foundation, may offer advantages through timeliness, enabling earlier interventions and strategic adjustments. While later predictions with a stronger data foundation may achieve greater accuracy, the added value of such precision must be balanced against the potential lag in responding to the identified delay. After all, this analysis should identify the optimal point in the ordering process where predictions can deliver actionable and sufficient results, ensuring that the effort invested in model development aligns with practical utility and maximized operational impact.

Nevertheless, research to generally improve algorithm prediction performance should be continued. An essential point to carry this out arises from the application of predictions in different industrial sectors (E), such as the consideration of the different characteristics of make-to-order and make-to-stock production processes. In MTO environments, orders are typically demand-driven and initiated by specific purchase requests, resulting in more dynamic and case-specific data. In contrast, MTS environments are based on automated purchasing processes, often regulated by long-term contracts, resulting in more complicated data patterns. Furthermore, complexity increases in multi-tier supply chains, requiring models that can adapt to divergent data sources and account for higher levels of uncertainty. These fluctuations underscore the need for robust predictive models to deliver equal performance in diverse industrial settings. In this context, further research should also conduct an analysis that shows the specific changes in prediction performance when using different data classes (A). The relevant publications have already shown that a higher number of different data classes leads to improved prediction performance, but without identifying the specific influence of individual data classes. It has also been shown that information such as location and environmental data, in addition to classic order or material data, delivers improved performance. In the future, however, it will be necessary to analyze the extent to which the advantages of combining different data classes apply and at which point an optimum is achieved. Furthermore, future research should also consider the integration of macroeconomic indicators, such as fuel prices or inflation rates, to better capture external influences and systemic risks in delivery delay prediction. In this context, future research should also examine the extent to which it is possible to establish guidelines for the comparability of data sets to better compare the results of individual publications. Also, as data sources and structures become increasingly diverse, future research may benefit from a structured maturity model to track the evolution of data usage in delivery delay prediction. A framework for such classification could consider various dimensions, including data richness (e.g., from internal-only to multi-source), temporal resolution, integration depth (e.g., static data vs. real-time), and accessibility. This concept may contribute to a systematic assessment of progress and identification of gaps in the practical use of data over time.

To fully utilize the relevant data in the prediction models, further research should also prioritize the development and use of regression algorithms (B) over classification algorithms, as these provide a significantly greater practical value for the direct prediction of delivery delays by determining a specific delay duration. At the same time, research should continue to focus on the development of such regression algorithms that can effectively process a larger number of features from different data classes. Real-time data such as environmental or location data, which may not be available from the ERP system, should be integrated into the prediction. However, before integrating such features into predictive models, a cost-benefit analysis is essential to determine whether the added complexity provides sufficient value. Another relevant topic on the research agenda concerns the advancement of the federated learning approach (H). This method offers a significant advantage for the development and improvement of prediction models by enabling the collective use of resources while ensuring data confidentiality and compliance with privacy regulations. These properties make it possible to divide the general further development of prediction models among different entities, whereby all entities benefit from the corresponding interaction without problems arising in the context of data security. Therefore, this approach results in an important objective for further research. The last point on the research agenda is the further development of classification algorithms regarding the use of class divisions to determine the length of delay compared to a classic yes/no prediction (G). These algorithms can approximate regression output when using delay-duration categories. Likewise, these models can offer some advantages, such as better interpretability, robustness to outliers, and suitability for threshold-based or rule-driven decision systems. However, further research is needed to evaluate whether these advantages offer significant added value in the specific context of predicting delivery delays and how they perform in practical industrial applications compared to regression algorithms. A summary of the research agenda is provided in the following Table 5.

Table 5
Research agenda for further developing the field of predicting delivery delays.

Prioritization	Further research need/recommendation
1	Integration of existing predictive algorithms into ERP systems to provide a direct benefit for the manufacturing industry.
2	Development of decision support systems to generate automated recommendations based on the executed prediction.
3	Execution of cost-benefit analyses to determine the optimal time of prediction and to determine the most economical quality of the prediction models.
4	Carrying out predictions in various industry sectors, especially in the differentiation between make-to-order and make-to-stock as well as corresponding multi-level supply chains to develop more robust models.
5	Conducting analyses to determine the effects of different data classes on the prediction performance to identify the optimal number and combination of different data classes.
6	Development of regression algorithms that can incorporate real-time data such as environmental data or location data into the prediction.
7	Further development and application of the federated learning approach to utilize synergy and economies of scale in the general further development of predictive algorithms.
8	Use of classification algorithms to predict the duration of the delay by categorizing it into different durations, as compared to a simple yes/no prediction.

In summary, the analysis of future research needs identified in the relevant publications also addresses RQ4 by highlighting frequently mentioned topics and identifying specific, uniquely mentioned approaches. Overall, the analysis reveals that a clear roadmap for the prediction of delivery delays has been lacking to date. Therefore, the developed agenda provides a structured foundation for advancing the research field in a targeted and efficient manner. To guide future research, the agenda prioritizes the practical integration of predictive models into ERP systems, the incorporation of real-time and multi-source data, and the development of DSS. As next steps, researchers should prioritize the practice-oriented topics such as implementing and

testing existing algorithms in industrial contexts and developing DSS, while addressing organizational and technical challenges to system integration. These challenges may include outdated IT infrastructures, limited data interoperability, or resistance to system-driven decisions in operating environments. To fully exploit the potential of predictive models in SCM and to turn theoretical insights into practical applications, future research must not only be methodologically innovative, but also actively address real-world constraints to enable sustainable and scalable implementation.

Advancing delivery delay prediction and implementing the outlined research directions will require strong interdisciplinary collaboration. The development and deployment of robust and usable prediction models depends not only on data scientists, but equally on supply chain experts with an industrial background, IT professionals, and user interface designers. Their combined expertise is essential to ensure that models are trained on meaningful data, integrated into existing systems, and translated into actionable insights. Future research initiatives should therefore actively promote cross-functional collaboration to fully realize the potential of predictive analytics in supply chain management.

6. Limitations

This paper has several limitations, which are discussed in the following section. A number of these limitations specifically relate to the literature selection process. First, the selection of literature databases may represent a potential limitation. While the ten databases selected encompass all major publishers, it is possible that relevant studies published by smaller or niche publishers were missed. Secondly, the exclusive use of English-language publications in the search process introduces a restriction. Although English is the leading language of scientific research, relevant research published in other languages might have been overlooked as a result.

Another limitation in the literature selection process results from the clustering. Although the clustering was based on a structured and validated approach, and supported by sample-based verification, a residual risk of misclassification or the omission of relevant publications cannot be entirely excluded. There is also a limitation related to the defined QC. While the applied criteria are comprehensive in their coverage of relevant thematic aspects, they do not incorporate formal quality assessment tools such as CASP or AMSTAR. The methodological quality was ensured exclusively by restricting the selection to peer-reviewed journal articles and conference proceedings, which guarantees a fundamental level of quality. However, this also constitutes a methodological limitation of the present SLR. Future reviews should enhance both the assessment of individual studies and the overall quality of reviews by systematically applying established quality evaluation frameworks.

A further limitation of the paper emerges from the prioritization of the research agenda. The prioritization of future research needs was based on expert assessments in workshops, which may have been influenced by subjective biases and the limited number of participants. Additionally, the results of this workshops reflect the perspective of German manufacturing companies and may not be directly transferable to other regions or industries.

7. Conclusion

This study has comprehensively examined and systematically reviewed the current state of research in the area of direct prediction of delivery delays. The conducted SLR identified the models, algorithms, and data used in existing literature, assessed their practical readiness, and revealed research gaps that served as the basis for the development of a research agenda. The literature search followed the established approach by Vom Brocke et al. [48], which was extended by the approaches of Cooper [50] and Moher et al. [54]. To select the identified literature, a clustering approach according to Weißer et al. [55] was used, supplemented by four quality criteria and a more extensive literature review. The content analysis and summary of the relevant publications was structured using a developed framework consisting of the five categories: forecast area, supervised learning, machine learning algorithm, data origin and data class.

The results show that the direct prediction of delivery delays is based almost equally on classification and regression algorithms. This highlights the diversity of approaches in current research and shows possible areas for optimization. Predominantly order and material data are used, with historical company data frequently forming the foundation. Although regression algorithms that provide concrete predictions about the delay duration offer significantly greater benefits for industrial applications, a considerable proportion of the publications rely on classification algorithms, even though regression can be performed using the same datasets. It also shows that the combination of more than two data classes, especially environmental or location-related supplier data, improves prediction quality.

Building upon the analysis, a research agenda for further research in this field was derived. The agenda systematically addresses the open questions and prioritizes them according to their relevance for practical application. In particular, the integration of existing prediction models in ERP systems for concrete use in companies represents one of the central research points. Moreover, the agenda can serve as a practical guide not only for further research, but also for the practical integration of prediction models into industrial applications.

The findings presented in this study offer several important implications across multiple domains. From a research perspective, the results contribute to a more structured understanding of the predictive models used to predict delivery delays and highlight the need to prioritize research towards regression-based approaches and comprehensive data integration. On an operational level, a key takeaway is the recommended integration of existing predictive models directly into companies' ERP systems, significantly enhancing their ability to proactively anticipate and mitigate delays. Finally, for supply chain decision makers, the development of integrated DSS is highlighted as essential. Such tools translate predictive results into actionable measures, improving companies' ability to effectively manage disruptions.

By systematically consolidating fragmented research and introducing a structured analytical framework, this study provides a clearer conceptual basis for future research. It advances the field by offering the first comprehensive research agenda tailored specifically to directly predicting delivery delays. In doing so, it shifts the focus from isolated methodological contributions to a more integrated and application-oriented research perspective.

CRediT authorship contribution statement

Fabian Steinberg: Writing – review & editing, Methodology, Conceptualization. **Carl René Sauer:** Writing – review & editing, Methodology, Conceptualization. **Maximilian Schütz:** Writing – review & editing, Conceptualization. **Norman Müller:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Peter Burggräf:** Supervision, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Appendix A

Table 6
Specific algorithms used in each of the different publications.

Document Title	Publication Year	Typ of Algorithm
A data mining approach to forecast late arrivals in a transhipment container terminal	2014	Classification and Regression Tree (CART)
A Hybrid Statistical Method for Accurate Prediction of Supplier Delivery Times of Aircraft Engine Parts	2015	Combined Hybrid Stepwise Regression and Multivariate Gamma Distribution
A novel machine learning model for predicting late supplier deliveries of low-volume-high-variety products with application in a German machinery industry	2023	Linear Regression (LR) Support Vector Regression (SVR) Decision Tree (DT) Random Forest (RF) Adaptive Boosting (AB) Gradient Boosting (GB) Artificial Neural Network (ANN)
Application of Machine Learning Algorithms for Order Delivery Delay Prediction in Supply Chain Disruption Management	2023	Logistic Regression (LogReg) K-nearest Neighbors (KNN) Random Forest (RF)
Development of Predictive Models for Order Delivery Risk in a Supply Chain: A Machine Learning Approach	2023	Logistic Regression (LogReg) Random Forest (RF) Naive Bayes (NB)
Federated machine learning for privacy preserving, collective supply chain risk prediction	2023	Artificial Neural Network (ANN) Convolutional Neural Network (CNN1D) Logistic Regression (LogReg)
Forecasting the time delay in delivery of pharmaceutical products	2022	Extra Trees (ET) Random Forest (RF)
Lead time forecasting with machine learning techniques for a pharmaceutical supply chain	2021	K-nearest Neighbors (KNN) Support Vector Regression (SVR) Random Forest (RF) Linear Regression (LR) Artificial Neural Network (ANN)
Novel Data Analytics Meets Conventional Container Shipping: Predicting Delays by Comparing Various Machine Learning Algorithms	2020	Support Vector Machines (SVM) Random Forest (RF) Artificial Neural Network (ANN) Support Vector Regression (SVR) Artificial Neural Network (ANN) Linear Regression (LR) Random Forest (RF)
On Bayesian forecasting of procurement delays: A case study	2006	Bayesian Networks (BN)
Predicting Purchase Orders Delivery Times Using Regression Models With Dimension Reduction	2018	Random Forest (RF) Linear Regression (LR)
Predicting supply chain risks using machine learning: The trade-off between performance and interpretability	2019	Support Vector Machines (SVM) Decision Tree (DT)
Predicting Delivery Outcome In Supply Chain Management Using Machine Learning: A Random Forest Classifier Approach	2024	Random Forest (RF)
Ship arrival prediction and its value on daily container terminal operation	2018	Random Forest (RF) Classification and Regression Tree (CART) Artificial Neural Network (ANN) Random Forest (RF)
Supply Chain Data Analytics for Predicting Delivery Risks Using Machine Learning	2023	Random Forest (RF) K-nearest Neighbors (KNN) Support Vector Machines (SVM) Logistic Regression (LogReg)
Supply chain data analytics for predicting supplier disruptions: a case study in complex asset manufacturing	2019	Random Forest (RF) Support Vector Machines (SVM) Logistic Regression (LogReg) K-nearest Neighbors (KNN) Linear Regression (LR)
Using a Fuzzy-Bayesian Approach for Predictive Analysis of Delivery Delay Risk	2022	Bayesian Networks (BN)
A Machine Learning Predictive Model for Shipment Delay and Demand Forecasting for Warehouses and Sales Data	2022	Naive Bayes (NB) Decision Tree (DT) K-nearest Neighbors (KNN) Artificial Neural Network (ANN)
Case study on delivery time determination using a machine learning approach in small batch production companies	2024	Support Vector Regression (SVR) Artificial Neural Network (ANN) Gradient Boosting (GB) Decision Tree (DT) Random Forest (RF) Linear Regression (LR)
Enhancing supply chain resilience: A machine learning approach for predicting product availability dates under disruption	2024	Linear Regression (LR) Random Forest (RF) Gradient Boosting (GB) Artificial Neural Network (ANN)
A neural network-based model for estimating the delivery time of oxygen gas cylinders during COVID−19 pandemic	2022	Artificial Neural Network (ANN) Support Vector Regression (SVR)

(*continued on next page*)

Table 6 (*continued*)

Document Title	Publication Year	Typ of Algorithm
Using MCDM Methods To Optimise Machine Learning Decisions For Supply Chain Delay Prediction: A Stakeholder-Centric Approach	2024	Linear Regression (LR) Decision Tree (DT) Random Forest (RF) Adaptive Boosting (AB)
A supervised machine learning approach to data-driven simulation of resilient supplier selection in digital manufacturing	2024	K-nearest Neighbors (KNN) Logistic Regression (LogReg)
Collaborative and integrated data-driven delay prediction and supplier selection optimization: A case study in a furniture industry	2024	Decision Tree (DT) Random Forest (RF) Gradient Boosting (GB)
Deep learning approaches to identify order status in a complex supply chain	2024	Random Forest (RF) K-nearest Neighbors (KNN) Artificial Neural Network (ANN) Support Vector Machines (SVM)
Machine Learning for Proactive Supply Chain Risk Management: Predicting Delays and Enhancing Operational Efficiency	2024	Linear Regression (LR) Random Forest (RF) Gradient Boosting (GB) Decision Tree (DT) Support Vector Machines (SVM) K-nearest Neighbors (KNN) Logistic Regression (LogReg) Decision Tree (DT)
Predicting late delivery in Supply chain 4.0 using feature selection: a machine learning model	2022	Random Forest (RF)
Predicting Delays for Truck Delivery Logistics: An Application of AI and ML	2024	Logistic Regression (LogReg) Random Forest (RF) Support Vector Machines (SVM) K-nearest Neighbors (KNN) Artificial Neural Network (ANN)
Predictive Analysis Methodology for Industrial Systems: Application in Supplier Delays Prediction	2022	Decision Tree (DT) Random Forest (RF) Naive Bayes (NB)

Group 1										
	A	B	C	D	E	F	G	H	Σ	Priorization
A	X	0	0	0	1	0	2	1	4	Prio 6
B	2	X	0	0	0	0	1	2	5	Prio 5
C	2	2	X	0	1	0	2	2	9	Prio 3
D	2	2	1	X	2	1	2	2	12	Prio 2
E	1	1	1	0	X	0	2	2	7	Prio 4
F	2	2	2	1	2	X	2	2	13	Prio 1
G	0	1	0	0	0	0	X	0	1	Prio 8
H	0	1	0	0	0	0	2	X	3	Prio 7

Fig. 17. Pairwise comparison matrix for group 1.

Group 2										
	A	B	C	D	E	F	G	H	Σ	Priorization
A	X	2	0	0	1	0	2	1	6	Prio 5
B	1	X	0	0	0	0	2	2	5	Prio 6
C	2	2	X	0	1	1	2	2	10	Prio 3
D	2	2	2	X	2	1	2	2	13	Prio 1
E	1	1	1	0	X	0	2	2	7	Prio 4
F	2	2	1	0	2	X	2	2	11	Prio 2
G	0	1	0	0	0	0	X	1	2	Prio 8
H	1	1	0	0	0	0	2	X	4	Prio 7

Fig. 18. Pairwise comparison matrix for group 2.

Group 3										
	A	B	C	D	E	F	G	H	Σ	Priorization
A	X	2	0	0	1	0	2	2	7	Prio 4
B	1	X	0	0	1	0	2	2	6	Prio 5
C	2	2	X	1	1	1	2	2	11	Prio 2
D	2	2	1	X	2	1	2	2	12	Prio 1
E	1	1	1	0	X	0	2	2	7	Prio 4
F	2	2	1	0	1	X	2	2	10	Prio 3
G	1	1	0	0	0	0	X	2	4	Prio 6
H	1	1	0	0	0	0	1	X	3	Prio 7

Fig. 19. Pairwise comparison matrix for group 3.

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